Cognitive Modeling

Complete Lecture Notes

Week 8: Midterm Review: The Whole Pipeline in One Case Study

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Chapter 1

Evaluation of Cognitive Models

1.1 Week 8: Midterm Review: The Whole Pipeline in One Case Study

Chapters 1–3 review · Lecture 1 h (integrated case study) · Lab slot: midterm exam (90 min) + project consultation (30 min)

1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

English	فارسی	توضیح کوتاه
formal model / formalize	فرمال کردن / مدل فرمال	نوشتن نظریه به شکل معادله یا الگوریتم
simulation	شبیه‌سازی	تولید داده‌ی مصنوعی با اجرای مدل
fit / fitting	برازش	تنظیم پارامترها تا مدل با داده جور شود
parameter recovery	بازیابی پارامتر	آیا برازش، پارامترهای معلوم شبیه‌سازی را برمی‌گرداند
likelihood	درست‌نمایی	احتمال داده به‌عنوان تابعی از پارامترها
criterion (c)	ملاک	آستانه‌ی تصمیم؛ نشان‌دهنده‌ی سوگیری پاسخ
least squares	کمترین مربعات	روش برازش با کمینه کردن مجموع مربع خطاها
cross-validation (CV)	اعتبارسنجی متقابل (cross-validation)	برازش روی بخشی از داده و آزمون روی بخش کنارگذاشته

English	فارسی	توضیح کوتاه
sensitivity (d′)	حساسیت	فاصله‌ی توزیع علامت و نویز
parameter	پارامتر	کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند
SSE / RMSD	ریشه‌ی میانگین / مجموع مربعات خطا مربع انحراف	معیارهای فاصله‌ی داده و پیش‌بینی
log-likelihood / NLL	منفی آن / لگاریتم درست‌نمایی	مقداری که بهینه‌سازها کمینه یا بیشینه می‌کنند
estimate / estimator	برآوردگر / برآورد	روش به / مقدار به‌دست‌آمده برای پارامتر دست آوردن آن
noise	نویز	تغییرات تصادفی در شواهد یا در داده
deviance (G^2)	انحراف (deviance)	معیار عدم برازش نسبت به مدل اشباع‌شده
AIC / BIC	بیزی / معیار اطلاعاتی آکائیکه	برازش به‌علاوه‌ی جریمه برای تعداد پارامتر
research design	طرح آزمایش	شرایط، تعداد کوشش‌ها و نمونه‌ها
confusion matrix	ماتریس درهم‌ریختگی (confusion matrix)	P(مدل انتخاب‌شده
hit	اصابت (hit)	پاسخ «بله» وقتی علامت وجود دارد
response time (RT)	زمان واکنش	فاصله‌ی ارائه‌ی محرک تا پاسخ
signal detection theory (SDT)	نظریه‌ی تشخیص علامت	مدلی که حساسیت را از سوگیری پاسخ جدا می‌کند
evidence	شواهد	اطلاعاتی که برای تصمیم جمع می‌شود
exponential / power function	توانی / تابع نمایی	دو شکل رایج برای منحنی فراموشی و یادگیری
training / test data	آزمون / داده‌ی آموزش	داده‌ی کنارگذاشته برای / داده‌ی برازش سنجش

1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Execute the complete modeling pipeline (formalize → simulate → fit → evaluate → recover → interpret) on a new problem without step-by-step guidance.
2. Select an appropriate likelihood, fitting method and evaluation method for a given dataset, and justify each choice.
3. Interpret a set of competing model fits, information criteria and recovery results, and write a defensible conclusion that states its limits.
4. Recall and apply the key formulas of Weeks 1–7 (the midterm formula sheet).

1.1.3 Motivation & big picture

Over seven weeks we built a toolkit piece by piece: formal models (Weeks 1–2), least squares and optimization (Week 3), maximum likelihood (Week 4), uncertainty and multiple participants (Week 5), cross-validation (Week 6), information criteria and recovery (Week 7). Each lab practiced one or two pieces. Real research needs all of them at once, in the right order, with judgment about what matters.

Today's lecture runs the entire pipeline on a new problem: does studying a word more often make it more *memorable* (higher sensitivity), and if so, how does sensitivity grow with repetitions? The case study is also the best preparation for the midterm, which asks you to make and justify the same kinds of decisions.

1.1.4 Lecture notes

In plain words This week puts all the pieces of the first half together on one new question: *does studying a word more often make it easier to recognize later?* We write the theory as an SDT model in which sensitivity d' depends on the number of repetitions. We propose three versions of that dependence (no effect, a smooth increase, any pattern), fit all three by maximum likelihood, check whether each misfits (G^2), compare them (LRT, AIC, BIC), check that the method works at this design (recovery), and finally interpret the winner in plain psychological language, including what the data *cannot* tell us. If you can follow this one example from start to end, you understand the first half of the course.

Suggested timing (60 min): concept map 10 · case study 40 · exam briefing 10.

1. Concept map of Weeks 1–7

Pipeline step	Question	Tools	Week
Formalize	What exactly does the theory claim?	Five components of a model; parameters as psychological constructs	1–2
Simulate	What can (and can't) the model do?	Forward simulation, analytic checks, prediction plots	1–2

Pipeline step	Question	Tools	Week
Fit	Which parameter values best account for the data?	SSE / least squares; NLL / MLE; grid, Nelder–Mead, L-BFGS-B; multi-start; bounds and reparameterization	3–4
Quantify uncertainty	How precise are the estimates?	Hessian (Wald) SEs, profile likelihood, bootstrap	5
Evaluate (absolute)	Does the model misfit beyond noise?	Residuals, G^2	2–4
Evaluate (relative)	Which model generalizes best?	CV, LRT, AIC/AICc/BIC, weights	6–7
Recover	Can the method work at this design?	Parameter recovery, model recovery (confusion and inversion matrices)	7
Interpret	What does it mean psychologically, and what can't we conclude?	Parameter meaning, design limits, aggregation artifacts	all

Decision guide for the likelihood:

Data	Natural noise model	Week
Counts out of n (recall, hits, false alarms)	Binomial	2, 4
Single binary choices	Bernoulli	4
Continuous, roughly symmetric, equal variance (e.g., block-mean RTs)	Gaussian (= least squares)	4–5

2. Case study: does repetition increase memory sensitivity?

Design. In a recognition-memory experiment, words are studied 1, 2 or 4 times. At test, 100 old words per repetition condition and 100 new words (lures) are presented, and the participant says "old" or "new". Because lures were never studied, there is one false-alarm rate, shared by all conditions, and three hit rates.

Formalization (SDT with an absolute criterion). New words produce evidence $X \sim \mathcal{N}(0,1)$; old words studied r times produce $X \sim \mathcal{N}(d'_r, 1)$; the participant says "old" if $X > \kappa$, with one criterion κ for all test items (they are intermixed, so the participant cannot know an item's condition). Then

$$H_r = \Phi(d'_r - \kappa), \qquad F = \Phi(-\kappa).$$

Three candidate models of how d' depends on repetitions:

Model	d'_r	k	Claim
$\mathcal{M}_0$ (no effect)	δ_0	2	Repetition does not change sensitivity
$\mathcal{M}_1$ (log-linear)	$\delta_0 + \delta_1 \log_2 r$	3	Each *doubling* of study adds the same sensitivity
$\mathcal{M}_2$ (free)	d'_1, d'_2, d'_4	4	Any pattern (saturated: 4 parameters for 4 rates)

$\mathcal{M}_0 \subset \mathcal{M}_1 \subset \mathcal{M}_2$, so all pairs are nested.

Likelihood. Binomial for each of the four counts (three hit counts, one false-alarm count).

The lab code below runs every step. Students run it at home as exam preparation; the lecture walks through the outputs.

Reading the math

- $H_r = \Phi(d'_r - \kappa)$ and $F = \Phi(-\kappa)$: the same SDT as Week 2, but the criterion is written as an absolute position κ on the evidence axis (measured from the center of the "new word" curve) instead of relative to the midpoint. New words: curve at 0, so F is the area right of κ, i.e. $\Phi(-\kappa)$. Old words studied r times: curve at d'_r, so $H_r = \Phi(d'_r - \kappa)$.
- One κ for all conditions: all test words are mixed together, so the participant cannot use a different threshold for words studied once and words studied four times.
- $\log_2 r$: the base-2 logarithm, so $\log_2 1 = 0$, $\log_2 2 = 1$, $\log_2 4 = 2$. In model $\mathcal{M}_1$, $d'_r = \delta_0 + \delta_1 \log_2 r$ means: δ_0 is the sensitivity after one study, and every *doubling* of study adds δ_1.
- Numbers from the worked example: $\hat{\kappa} = 1.04$ gives $F = \Phi(-1.04) = 0.15$ ✓. With $\hat{d}'_1 = 0.82$: $H_1 = \Phi(0.82 - 1.04) = \Phi(-0.22) = 0.41$, close to the observed 0.40 ✓.

For an engineer The pipeline of the concept map is the system identification workflow you know: choose candidate model structures, estimate parameters (ML), validate (residual tests $\approx G^2$), select the model order (LRT/AIC/BIC $\approx$ GLRT/AIC/MDL), and verify the procedure on simulated data with known truth (recovery). Two differences deserve emphasis in class: the "system" is a person whose parameters are themselves the scientific result, and the amount of data is tiny by engineering standards (400 binary responses), so uncertainty and recovery matter much more.

For the instructor

- Use the concept-map table as the backbone of the session. For each row, point to the corresponding output of the case study (e.g. "Evaluate (absolute)" $\rightarrow$ the G^2 column). Students see that every tool has a place.
- Ask the class to interpret before you do. Show the results table without the interpretation paragraph and let small groups write two sentences: "what do we conclude, and what can't we conclude?" Then compare with the text. This is exactly the skill the midterm tests.
- Exam briefing: tell students that the exam rewards *justified decisions* (which likelihood, which criterion, why recovery matters) more than computation. Show one past-style question from the "Check your understanding" section.
- Project consultation (lab slot): ask each team three questions: What is your cognitive question? Which parameter or model comparison answers it? Can your design recover it? Most weak proposals fail on the third.

1.1.5 Worked example

The case-study results for one simulated participant (true model $\mathcal{M}_1$ with $\delta_0 = 0.8$, $\delta_1 = 0.6$, $\kappa = 1.0$), from the code in the lab section. Numbers are rounded.

Step 1: data.

Condition	Old words	"Old" responses	Rate
$r = 1$	100	40	0.40
$r = 2$	100	66	0.66
$r = 4$	100	80	0.80
New (lures)	100	15	0.15

Step 2: fit and evaluate (MLE, $N = 400$ responses):

Model	k	NLL	G^2 (df, p)	AIC	BIC	w_{AIC}	w_{BIC}	$\hat{d}'_{1,2,4}$	$\hat{\kappa}$
$\mathcal{M}_0$ none	2	27.27	35.55 (2, < .001)	58.54	66.52	0.00	0.00	1.34, 1.34, 1.34	1.04
$\mathcal{M}_1$ log-linear	3	9.77	0.54 (1, .46)	25.53	37.51	0.67	0.94	0.82, 1.37, 1.93	1.04
$\mathcal{M}_2$ free	4	9.50	0 (saturated)	26.99	42.96	0.33	0.06	0.78, 1.45, 1.88	1.04

Likelihood-ratio tests: $\mathcal{M}_0$ vs. $\mathcal{M}_1$: $\Lambda = 35.01$, df = 1, $p < .001$; $\mathcal{M}_1$ vs. $\mathcal{M}_2$: $\Lambda = 0.54$, df = 1, $p = .46$.

Recovery at this design (lab code, second block): $\mathcal{M}_1$'s parameters are well recovered ($r = 0.89$–0.93), and BIC identifies the generating model in 87–98% of simulations.

Step 3: interpret. Repetition clearly increases sensitivity: the no-effect model misfits badly ($G^2 = 35.5$, $p < .001$) and is rejected by every criterion. The log-linear model fits as well as the saturated model (LRT $p = .46$) with one parameter fewer, and both AIC and BIC prefer it. Its estimate $\hat{\delta}_1 \approx 0.55$ means each doubling of study adds roughly half a d' unit. Recovery results show that, at this design, both the parameters and the model choice can be trusted. Caveats: this is one simulated participant; with only three repetition levels, a log-linear trend cannot be distinguished from many other increasing curves (e.g., a power function of r); and a single criterion κ assumes the participant cannot tell how often an item was studied before responding.

1.1.6 Lab (2 hours): midterm exam + project consultation

Midterm exam (90 minutes, in the lab slot)

- Format: written, closed book. A formula sheet (below) is provided. No laptops: questions involve reading short code fragments and interpreting outputs, not writing code from scratch.
- Coverage: Weeks 1–7 (Chapters 1–3), with the emphasis on *understanding and interpretation* (the course's specific goal 2).
- Question types: (1) short conceptual questions; (2) one calculation by hand (e.g., d' and c, a likelihood or NLL, AIC/BIC and weights); (3) reading code and predicting or explaining its output; (4) interpreting a fit table, a residual plot or a recovery/confusion matrix; (5) one short design question ("what would you change, and why?").
- Weight: 3 of 20 points.
- The midterm exam itself, with model answers, is in the course assessments (Stage 3 materials).

Formula sheet provided at the exam:

$$d' = \Phi^{-1}(H) - \Phi^{-1}(F), \quad c = -\tfrac{1}{2}[\Phi^{-1}(H) + \Phi^{-1}(F)], \quad H = \Phi(d'/2 - c), \quad F = \Phi(-d'/2 - c)$$

$$f_{\text{exp}}(t) = Ae^{-\lambda t}, \quad f_{\text{pow}}(t) = A(1+t)^{-d}, \quad P_c^{\text{RW}} = \frac{1-r^z}{1-r^a},\ r = \frac{1-v}{1+v}, \quad \mathbb{E}[T] = \frac{aP_c - z}{v}$$

$$\text{SSE} = \textstyle\sum_i (y_i - \hat{y}_i)^2, \quad \text{RMSD} = \sqrt{\text{SSE}/N}$$

$$\ell(\boldsymbol{\theta}) = \textstyle\sum_i \left[\ln \binom{n_i}{y_i} + y_i \ln f_i + (n_i - y_i)\ln(1 - f_i)\right], \quad \text{NLL} = -\ell$$

$$G^2 = 2(\ell_{\text{sat}} - \hat{\ell}), \quad \Lambda = 2(\hat{\ell}_1 - \hat{\ell}_0) \sim \chi^2_{\Delta k}$$

$$\text{SE} \approx \sqrt{\text{diag}(\text{H}^{-1})}, \quad \text{Wald CI: } \hat{\theta} \pm 1.96\,\text{SE}, \quad \text{profile CI: } 2(\text{NLL}_{\text{prof}} - \text{NLL}_{\text{min}}) \le 3.84$$

$$\text{AIC} = 2\,\text{NLL} + 2k, \quad \text{AICc} = \text{AIC} + \tfrac{2k(k+1)}{N-k-1}, \quad \text{BIC} = 2\,\text{NLL} + k\ln N, \quad w_i = \tfrac{e^{-\Delta_i/2}}{\sum_j e^{-\Delta_j/2}}$$

Critical values: $\chi^2_{1,.95} = 3.84$, $\chi^2_{2,.95} = 5.99$, $\chi^2_{3,.95} = 7.81$. Quantiles: $\Phi^{-1}(0.84) \approx 1.0$, $\Phi^{-1}(0.975) = 1.96$. Logs: $\ln 2 = 0.693$, $\ln 10 = 2.303$, $\ln 100 = 4.605$, $\ln 400 = 5.991$.

Project consultation (30 minutes, after the exam)

Teams meet the instructor or TA for 5–10 minutes each to discuss their project idea before the proposal deadline (start of Week 9). Bring a one-paragraph description of your question, model and data.

Self-study: the case study in code (run before the exam)

Code · کد: `code/week08.py`, cell 1 · سلول 1

Code · کد: `code/week08.py`, cell 2 · سلول 2

Expected output (first block):

```
hits per condition: [40 66 80]  false alarms: 15
           k     NLL      G2  df  p(G2)     AIC     BIC
           ↪  d'_1,2,4  kappa  w_AIC  w_BIC
model
M0_none    2  27.269  35.549   2  0.000  58.538  66.521  [1.34, 1.34,
 ↪  1.34]   1.04  0.000  0.000
M1_loglin  3   9.766   0.543   1  0.461  25.533  37.507  [0.82, 1.37,
 ↪  1.93]   1.04  0.674  0.938
M2_free    4   9.495   0.000   0    NaN  26.990  42.956  [0.78, 1.45,
 ↪  1.88]   1.04  0.326  0.062

LRT M0 vs M1: 35.01 (df 1, p = 0.0000);  M1 vs M2: 0.54 (df 1, p = 0.461)
```

Expected output (second block, about 1 minute):

```
M1 recovery delta0 : r = 0.895, RMSE = 0.181
M1 recovery delta1 : r = 0.930, RMSE = 0.113
M1 recovery kappa  : r = 0.887, RMSE = 0.142

BIC model recovery, P(selected | generated):
           M0_none  M1_loglin  M2_free
M0_none       0.98       0.02     0.00
M1_loglin     0.05       0.93     0.02
M2_free       0.10       0.03     0.87
```

Note on the $\mathcal{M}_2$ generator in model recovery: its three d' values are drawn so that d'_4 lies *between* d'_1 and d'_2, a non-monotone pattern that $\mathcal{M}_1$ cannot produce. Otherwise $\mathcal{M}_2$ data would often be perfectly consistent with $\mathcal{M}_1$, and "confusions" would be meaningless.

Self-check questions on the case study (discussed in the lecture): 1. Why is there only one false-alarm rate, and how does that change the SDT model compared with Week 2? 2. Which model does each criterion favor, and do the LRTs agree? 3. Is δ_1 recoverable at this design? Is $\mathcal{M}_1$ distinguishable from $\mathcal{M}_0$ and $\mathcal{M}_2$? 4. Write the one-sentence psychological conclusion, with its main caveat.

1.1.7 Connection to AI & LLMs

Chatbots as study partners: trust, but verify. Many students use LLMs to explain concepts and check exam answers. That can help, but LLMs can produce fluent, confident and wrong calculations. Treat a chatbot's answer as you would a model's prediction: check it against something you can compute. The code below checks a pre-made chatbot answer to midterm question B3 (AIC for three models). The chatbot used $2\,\mathrm{NLL} + k$ instead of $2\,\mathrm{NLL} + 2k$.

Code · کد: `code/week08.py`, cell 3 · سلول 3

Expected output:

```
M1: chatbot 222.0, correct 224.0  WRONG
M2: chatbot 218.0, correct 221.0  WRONG
M3: chatbot 218.8, correct 223.8  WRONG
Akaike weights, chatbot: [0.07 0.55 0.37]  correct: [0.15 0.68 0.17]
```

The chatbot happens to pick the right winner ($\mathcal{M}_2$), but every number is wrong, and it doubles the support for the most complex model. A grader would give little credit, and in a real analysis the error would favor overfitting.

Try it (no API needed). - ⋆ Which error in the formula produces exactly the chatbot's numbers? What would BIC look like with the same error? - ⋆⋆ Write three small "test cases" (inputs with known answers) for the SDT formulas of Week 2 that you could use to check any answer, from a chatbot or a classmate. - Reminder: no AI tools are allowed in the exam itself, and the course policy on AI use for assignments is in `assessments/README.md`.

1.1.8 Common mistakes & misconceptions (top 10 from Weeks 1–7)

1. Treating a good fit as evidence that a model is true (Weeks 3, 6).
2. Forgetting the minus sign: optimizers minimize the negative log-likelihood (Week 4).
3. Not clipping probabilities, so $\ln 0$ crashes the fit (Week 4).
4. Trusting `success=True`, a single start, or a bounded Nelder–Mead that has collapsed against a bound (Weeks 3–4).
5. Least squares for counts or choices without thinking about the noise model (Weeks 3–4).
6. Reporting estimates without uncertainty, or using Wald intervals near bounds (Week 5).
7. Drawing conclusions about individuals from averaged curves (Week 5).
8. Leaking test data into training in cross-validation (Week 6).
9. Comparing non-nested models with an LRT, or reading AIC values on their own instead of as differences (Week 7).
10. Interpreting parameters or model comparisons without a recovery study at the actual design (Week 7).

1.1.9 Check your understanding (practice questions in exam style)

1. (Concept) Explain in three sentences why the exponential-with-asymptote model fitted the Week 2 data better than the power model, although the power model generated the data. Answer The asymptote model has an extra parameter, so it can adapt to the particular noise of the sample, here the flat-looking long-delay points. Better training fit is guaranteed for more flexible models and is not evidence of truth. Generalization measures (CV, AIC/BIC) and recovery studies are needed to judge it.

2. (Calculation) An observer has $H = 0.84$ and $F = 0.16$. Compute d' and c (use $\Phi^{-1}(0.84) \approx 1.0$). Answer $d' = 1.0 - (-1.0) = 2.0$; $c = -\frac{1}{2}(1.0 + (-1.0)) = 0$, so the observer is unbiased.

3. (Calculation) Two models are fitted to $N = 100$ binary responses. Model A: $k = 2$, NLL = 60.0. Model B: $k = 4$, NLL = 57.5. Compute AIC and BIC for both. Which does each criterion prefer? ($\ln 100 = 4.605$) Answer AIC: A = 124.0, B = 123.0, so AIC prefers B (slightly). BIC: A = 120 + 9.21 = 129.21; B = 115 + 18.42 = 133.42, so BIC prefers A. The criteria disagree because BIC penalizes each parameter by 4.6 instead of 2.

4. (Code reading) What is wrong with this NLL function, and what will the optimizer do?

> Code · کد: `code/week08.py`, cell 4 · سلول 4

Answer (i) It returns the log-likelihood, not its negative, so `minimize` will drive the parameters toward the *worst* fit (log-likelihood toward $-\infty$). (ii) `p` is not clipped, so predicted values of 0 or above 1 give −inf or nan. Fix: `return -np.sum(binom.logpmf(y,n,np.clip(p,EPS,1-EPS)))`.

5. (Interpretation) A recovery study gives $r(\text{true}, \text{recovered}) = 0.95$ for parameter A and 0.35 for parameter B. What do you conclude about a paper reporting group differences in B at this design? Answer B is poorly recoverable at this design: estimates are dominated by noise or trade-offs, so group differences in estimated B are not interpretable. The paper should improve the design or not interpret B.

6. (Interpretation) A BIC confusion matrix has P(selected = power | generated = asymptote) = 0.58. Your real data's BIC winner is the power model. What should you write? Answer That the power model is preferred by BIC but, at this design, BIC often selects the power model for data generated by the asymptote model, so the evidence does not discriminate between them well. Report the inversion probabilities, AIC/CV results, and recommend a design with longer delays.

7. (Design) You want to know whether a drug changes caution (boundary a) or evidence quality (drift v) in the random walk model. What data must you collect, and why is accuracy alone not enough? Answer Choices and RTs (ideally whole RT distributions) under drug and placebo. Both a larger a and a larger v raise accuracy, but a slows responses while v speeds them, so only the joint pattern separates them (Week 1).

8. (Concept) Why must cross-validation folds for the forgetting data be formed within intervals if we want to test prediction of new trials? Answer So that every training set contains all retention intervals (conditions), and the test asks about new items in known conditions. Leaving out whole intervals would test extrapolation to new conditions, which is a different generalization claim.

1.1.10 Assignment

- A3 (model comparison and recovery) is due at the start of the Week 9 lecture.
- Project proposal is due at the start of the Week 9 lecture (max. 2 pages; see Week 7).
- No new assignment this week. Prepare for the midterm by running the case study above and answering the practice questions without looking at the answers.

1.1.11 Further reading

- Farrell & Lewandowsky (2018), chapters 2–5 and 10. Reread the chapter summaries as a review of the whole first half of the course.
- Lee & Wagenmakers (2014), chapter 11 (SDT). Read the model description as preparation for Week 9, where we start treating parameters as random variables with priors.

Preview for Week 9: the second half of the course turns Bayesian. Instead of a single best estimate plus a confidence interval, we will compute the full probability distribution of the parameters given the data.